

INNOVA JUNIOR COLLEGE  
JC2 PRELIMINARY EXAMINATION  
in preparation for General Certificate of Education Advanced Level  
**Higher 2**

CANDIDATE  
NAME

**MARK SCHEME**

CLASS

INDEX NUMBER

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**BIOLOGY**

**9648/03**

Paper 3 Applications Paper

**25 August 2016**

**2 hours**

Additional Materials: Answer Paper  
Cover Page

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**READ THESE INSTRUCTIONS FIRST**

Write your name and class on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [ ] at the end of each question or part question.

| For Examiner's Use |    |
|--------------------|----|
| 1                  | 16 |
| 2                  | 13 |
| 3                  | 11 |
| 4                  | 12 |
| 5                  | 20 |
| Total              | 72 |

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This document consists of **16** printed pages.



**Section A**Answer **all** questions.

- 1** Diabetes mellitus is a disease that results in the inability to regulate one's blood glucose levels, either due to lack of insulin or insensitivity to insulin.

Injection of insulin before meals helps patients to keep blood glucose within set point. Insulin used to be purified from animal sources but genetic engineering has now provided an alternative source.

- (a)** State **two** advantages of genetically engineered insulin compared to those obtained from animal sources.

**1. religious considerations**

*insulin may be purified from animals that violate religious practices e.g. Muslims cannot use porcine insulin;*

**2. insulin from animal sources may cause allergy rxn in patients**

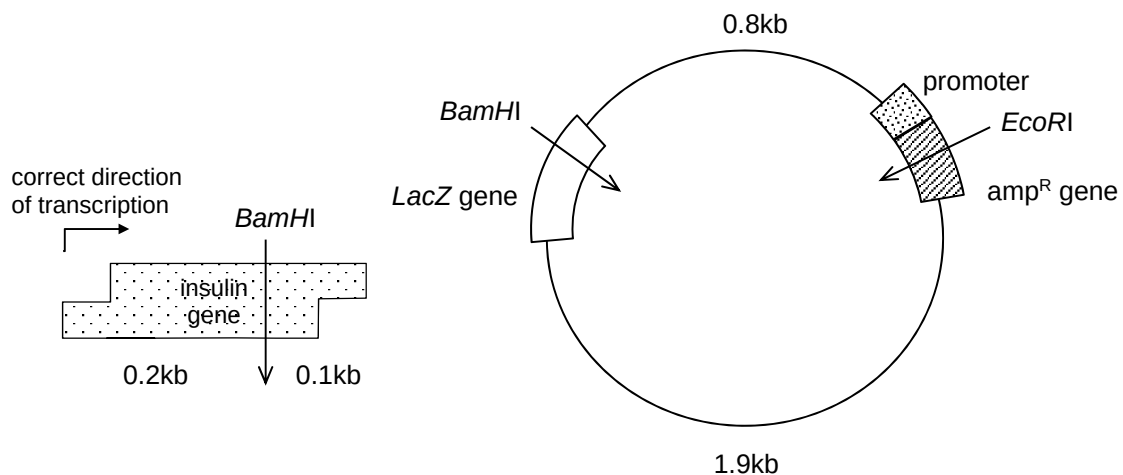
*GE source is identical to human insulin as it is transcribed & translated from human insulin gene;*

**3. can be produced in larger quantities**

*to meet high demands; (ANY 2)*

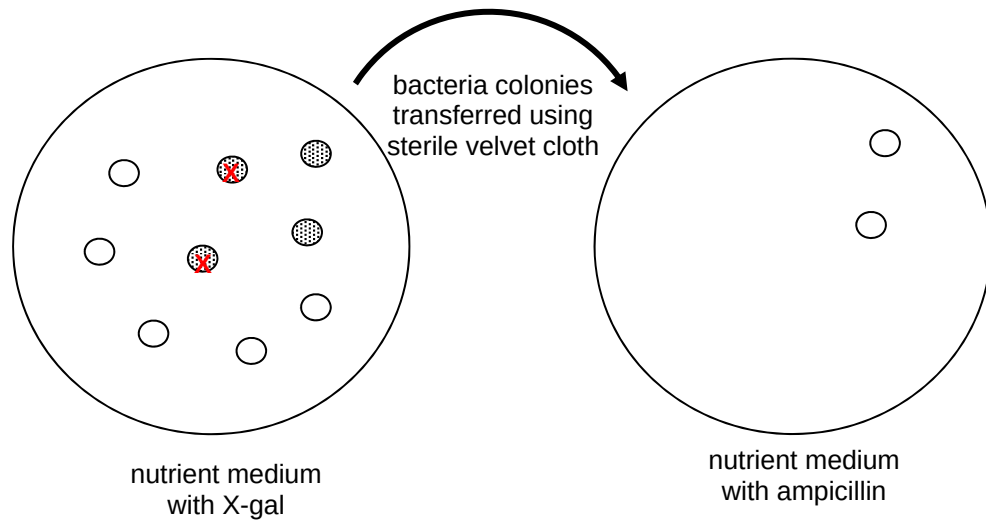
[2]

Human insulin is isolated using *EcoRI* from the cDNA library and ligated into plasmid shown in Fig. 1.1. The number of base pairs between the restriction recognition sites is shown.



**Fig. 1.1**

Plasmids were introduced into bacteria host cells and grown on nutrient medium containing X-gal. A replica plate was made by transferring bacteria colonies from the first plate to nutrient medium containing ampicillin. Colonies growing in the two plates are shown in Fig. 1.2.



**Fig. 1.2**

Legend

- blue colony
- white colony

(b) (i) State **one** possible bacteria host cell.

***Escherichia coli***

[1]

(ii) Explain how plasmids were introduced into host cells.

**1. (artificial) transformation**

***via heat shock;***

**2. in cold  $\text{CaCl}_2$**

***to induce competence / reduce repulsion btw mem & DNA;***

**3. in  $42^\circ\text{C}$  for 1 – 2 min**

***to cause temporary poration in mem;***

**4. back into  $0^\circ\text{C}$**

***to close pores to prevent plasmid from escaping;***

[4]

(c) (i) On Fig. 1.2, label the recombinant colony/colonies with an 'X'. [1]

(ii) Explain how you arrive at your answer in (c)(i).

**1. successfully transformed bact has plasmid intact lacZ gene**

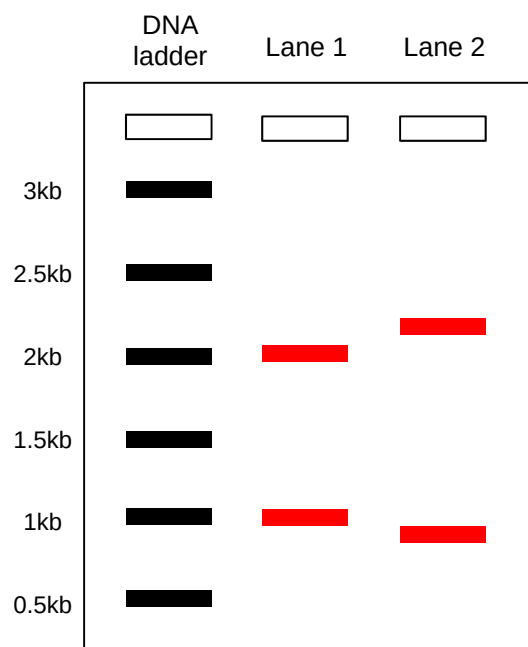
**coding for  $\beta$ -galactosidase which converts X-gal  $\rightarrow$  blue cpd thus are blue;**

**2. transformed bact with recombinant plasmid will have disrupted amp<sup>R</sup> gene / have undergone insertional inactivation**

**thus recombinant bact are blue colonies in the 1<sup>st</sup> plate that does not grow in replica plate;** [2]

The insulin gene was found to be inserted into plasmids in two possible orientations. To determine if the insulin gene was inserted correctly, recombinant plasmids were isolated, digested with *Bam*HI and separated by gel electrophoresis.

Fig. 1.3 shows the outline of the electrophoresed gel.



**Fig. 1.3**

(d) On Fig. 1.3, illustrate the bands found if plasmids with insulin gene inserted

(i) correctly was loaded in lane 1, [1]

(ii) wrongly was loaded in lane 2. [1]

(e) Describe how gel electrophoresis is used to separate the restriction fragments.

1. *restriction fragments are loaded in wells of agarose gel near cathode*  
-----  
*gel submerged in buffer with ions to maintain pH & conduct electricity;*  
-----
2. *DC / voltage applied aX gel*  
-----  
*-vely charged DNA migrates towards anode;*  
-----
3. *with gel acting as molecular sieve that impedes movement of DNA*  
-----  
*at speeds inversely proportional to fragment length;*  
-----
4. *fragments of the same length would be found at the same region*  
-----  
*forming a band;*

[4]

[Total: 16]

2 X-linked Severe Combined Immunodeficiency (X-SCID) is a recessive genetic disorder. Diseased individuals suffer from compromised immune system due to malfunctioning lymphocytes.

(a) (i) Describe the cause and explain the symptoms of X-SCID.

1. *mutation in gene on X chromosome*  
-----  
*coding for  $\gamma_c$  chain in IL2;*  
-----
2. *unable to bind cytokines / interleukins*  
-----  
*due to  $\Delta$  in 3D config;*  
-----
3. *unable to  $\oplus$  cell division & proliferation of B & T lymphocytes*  
-----  
*lack of / malfunctioning WBC  $\Rightarrow$  immune sys is compromised;*

[3]

(ii) Explain why males are more likely to suffer from the disease compared to females.

*males only have 1 copy of X chromosome  $\Rightarrow$  inheritance of a single copy of the recessive allele results in the disease;*

*Accept alternative argument of females having 2 X chromosome*

[1]

X-SCID can be treated by ex vivo gene therapy shown in Fig. 2.1.

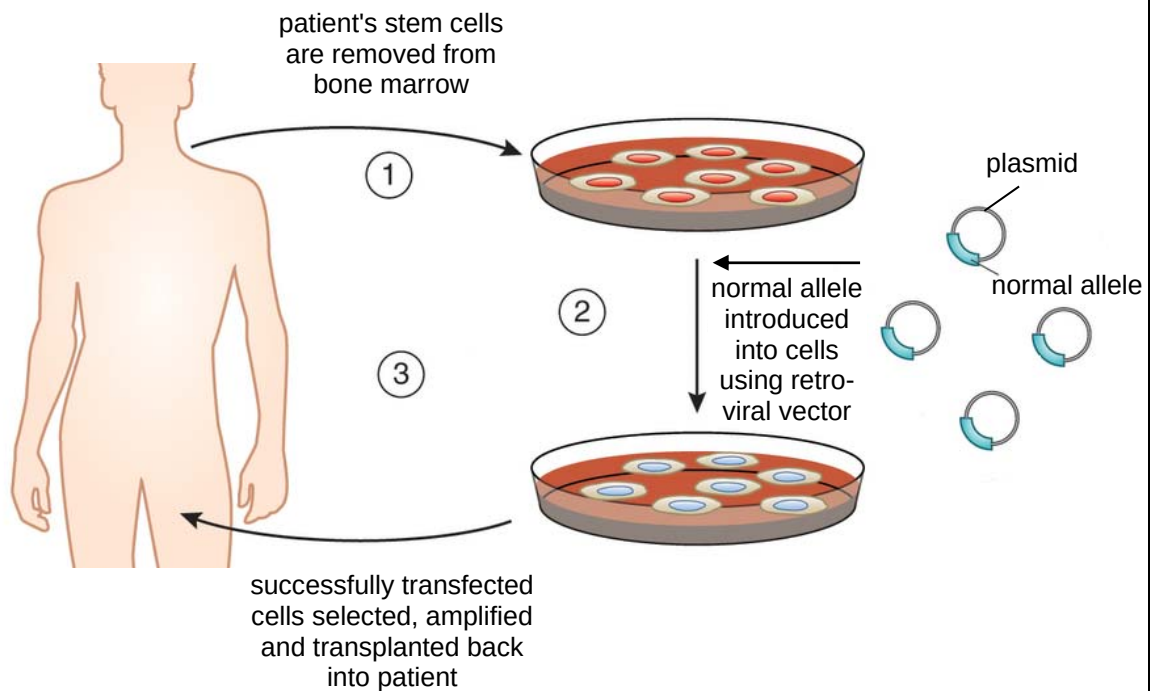


Fig. 2.1

- (b) (i) State **two** reasons why the ex vivo approach was used instead of *in vivo*.
- allows selection of successfully transfected cells;**
  - allows cells to be amplified in culture to ↑ nos. before transplanting**  
 ↑ chance of SC successfully residing in bone marrow;
  - allows cells to be screened for insertional mutagenesis;** [2]
- (ii) Suggest why the normal allele was ligated into a plasmid before introduction into cells.
- linear fragments of DNA are easily digested by cellular enz**  
 after intro into cells; [1]
- (iii) Suggest how successfully transfected cells were selected.
- DNA from some cells of the colony isolated**  
 subj to restriction digestion;
  - restriction fragments separated by gel electrophoresis**  
 to check for presence of transgene; [2]

Normal lymphocytes were detected in some patients' blood for several months following gene therapy. Some of the patients who underwent this treatment were subsequently diagnosed to suffer from leukaemia (cancer of blood cells).

- (c) (i) Explain why normal lymphocytes were detected in patients' blood for several months following gene therapy.

**1. normal allele is integrated into SC genome**

*stably propagated to daughter cells when SC divides;*

**2. transgene is present and expressed in new lymphocytes**

*allow maturation and proliferation of normal lymphocytes;*

[2]

- (ii) Explain how gene therapy could have led to the development of leukemia in other patients.

**1. due to insertional mutagenesis**

*where transgene is inserted in e.g. TSG;*

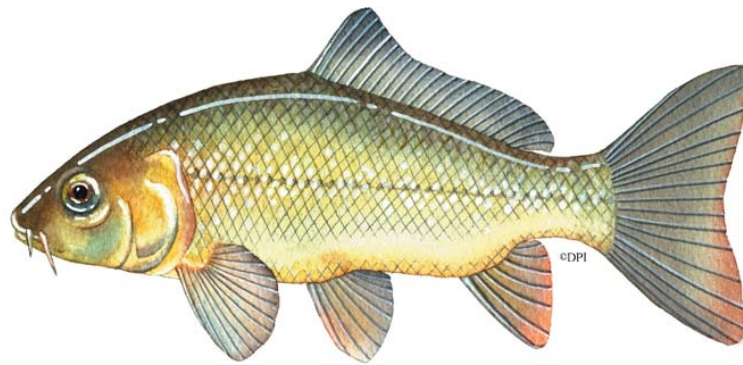
**2. leading to LOF mutation**

*causing SC to undergo uncontrolled proliferation to give too many WBCs;*

[2]

[Total: 13]

- 3 The carp, shown in Fig. 3.1, is a large freshwater fish native to central Asia. It was introduced to Australia by humans in the mid-1800s, where it subsequently became an aquatic pest.



**Fig. 3.1**

- (a) Suggest why the carp became a pest when it was introduced into Australia.

**1. no natural predators**

*thus reproduce at rapid rate; OR*

**2. is a large fish**

*able to outcompete local spp for resources / food;*

[1]

Scientists tried to control the carp by using gene technology. They inserted a gene into carp embryos that prevents the production of an enzyme, aromatase. Aromatase catalyses the conversion of testosterone into oestrogen. The inserted gene causes female embryos to develop as males.

The inserted gene produces mRNA with a base sequence that is complementary to the base sequence of the mRNA produced by the aromatase gene. Table 3 shows part of the base sequence of the aromatase gene. It also shows part of the base sequence of the mRNA produced by the inserted gene.

**Table 3**

|                                      | Base sequence |          |          |          |          |          |
|--------------------------------------|---------------|----------|----------|----------|----------|----------|
| Template strand of aromatase gene    | T             | G        | <b>G</b> | <b>C</b> | A        | <b>T</b> |
| mRNA transcribed from aromatase gene | <b>A</b>      | <b>C</b> | <b>C</b> | <b>G</b> | <b>U</b> | <b>A</b> |
| Template strand of inserted gene     | <b>A</b>      | <b>C</b> | <b>C</b> | <b>G</b> | <b>T</b> | <b>A</b> |
| mRNA transcribed from inserted gene  | <b>U</b>      | <b>G</b> | G        | C        | <b>A</b> | U        |



(b) Complete Table 3 to show the base sequences in the mRNAs transcribed and the inserted gene. [3]

(c) Describe how the inserted gene was introduced into the carp embryos.

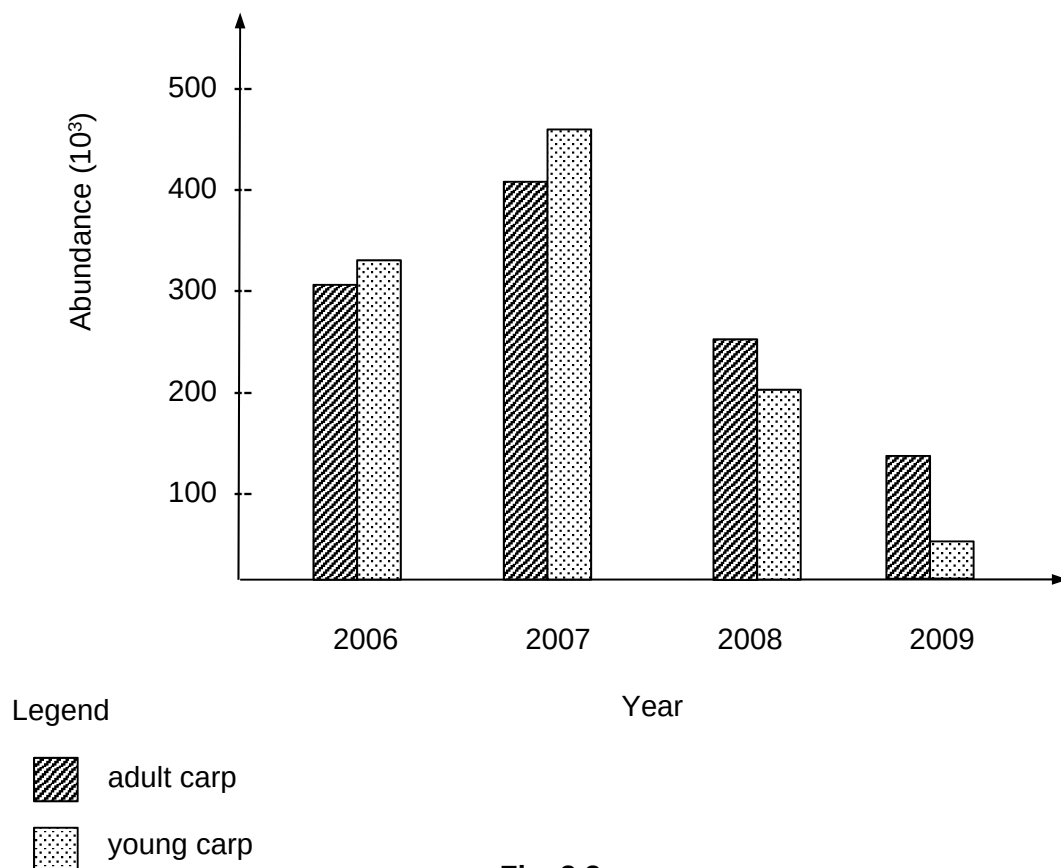
1. *by particle bombardment;*

2. *gene to be inserted is ligated into plasmids & coated on tungsten beads;*

*and shot at high speed into embryo cells;*

[2]

The genetically modified (GM) carp were released into the environment in 2006. Fig. 3.2 shows the effect on wild carp populations for the subsequent years.



**Fig. 3.2**

- (d) With reference to Fig. 3.2 and information on the inserted gene, explain the effect of releasing GM carp on carp populations.

1. *initial ↑ in 2007 due to more carp available for repro*

*both adult & young carp ↑ from 300,000 to 400,000 & 330,000 to 450,000 respectively;*

2. *popn of both adult & young carp ↓ from 2008 – 2009*

*to 150,000 for adult & 50,000 for young carp;*

3. *as GM carp mating with carp in wild & pass down inserted gene to offspring*

*inserted gene is transcribed into anti-sense mRNA;*

4. *that compl bp with aromatase mRNA (forming a duplex)*

*that ⊗ (ribosome binding thus) translation into aromatase;*

5. *testosterone is ⊗ converted into oestrogen ➡ all offspring develop into males*

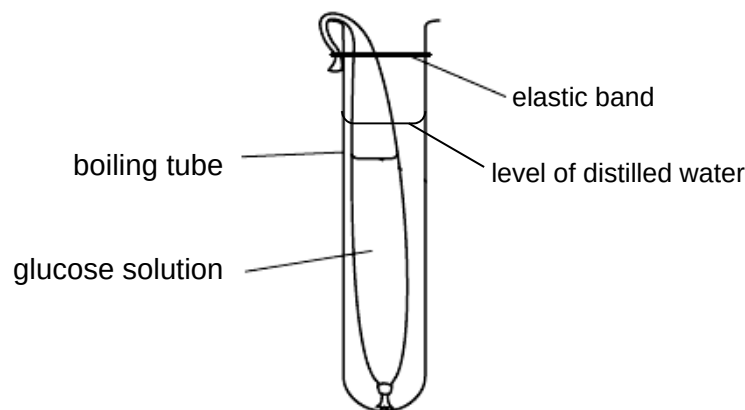
*➡ ↓ of no. of females in the popn thus ↓ repro rate;*

[5]

[Total: 11]

#### 4 Planning question

Visking tubing is a selectively permeable membrane that allows small molecules like glucose to pass through. Fig. 4.1 shows how a Visking tubing can be set up to investigate diffusion rates.



**Fig. 4.1**

Using this information and your own knowledge, design an experiment to investigate the effect of concentration on rate of glucose diffusion.

You must use:

- 10% glucose solution,
- distilled water,
- Benedict's solution,
- Visking tubing,
- boiling tube,
- elastic band.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- syringes,
- Bunsen burner, tripod stand.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

**Theoretical background [2m]**

1. define diffusion ref to movement from region of higher to lower conc., conc. gradient, passive process
2. state hypothesis ref to faster rate when [glc] is higher in Visking tubing, steeper conc. gradient, until equilibrium with no net movement

*ER: Some candidates think glucose diffusion is dependent on water potential rather than solute conc.*

**Independent variable [1m]**

3. conc. of glucose, at least 5 conc. of regular interval & wide range, vol of 10% glucose & distilled water used for dilution shown in dilution table (either simple or serial)

*ER: Some attempted to describe vol used in prose resulting in confusing phrasing. Others excluded final glucose conc., an important info, in the table.*

**Dependent variable [1m]**

4. amount of glucose diffused into distilled water measured by amount & colour of ppt in Benedict's test / time of 1<sup>st</sup> ppt appearance

**Benedict's procedure must be feasible to score this mark**

*ER: Majority of the candidates are not able to describe Benedict's test correctly either adding too little Benedict's soln e.g. 2 – 3 drops or not placing samples in boiling water bath.*

*Many candidates stated colour of ppt as data to record, failing to understand that amount of ppt also indicates glucose conc.*

*Candidates also did not think through their answers coherently and examiners wonder how one can calculate an average of colours in the results table.*

**Controlled variable (any 2) [2m] rationale must be stated**

5. temperature, use of water bath at specified temp ® **direct heating with Bunsen flame**

6. time allowed for diffusion

7. vol of glucose in Visking tubing

**Procedure [2m]**

8. coherent sequence of steps e.g. sequence of adding solns (® set up according to Fig. 1), starting stopwatch

9. rinsing Visking tubing, vol distilled water > glucose, removing sample for Benedict's test, time 1 tube at a time etc.

*ER: Majority of the candidates did not understand the basic principle of conducting expt. Many candidates heated the solutions while diffusion is occurring, being totally oblivious to the obvious fact that diffusion rates would be affected by this setup.*

*Some candidates listed temperature as a controlled variable and yet still heated the Visking tubing, showing a lack of understanding of what they are writing.*

*Others decided to record the time taken for brick red ppt to appear even for control setups.*

**Reliability [1m]**

10. replicates & repeats (**only if procedure is largely feasible**)

**Results [1m]**

11. table of results

*describe means of deciphering conc. of glucose in distilled water*

**® colour of ppt as results as average of replicates cannot be calculated nor plotted in graph**

**Graph [1m]**

12. plots theoretical graph of initial glucose conc. vs conc. of glucose diffused into distilled water / time / amt of ppt

**Safety [1m] (any 1)**

**13. states hazard, risk, precaution**

**e.g. hot water from boiling waterbath may cause scalding, handle waterbath with cotton gloves**

**e.g. Benedict's solution may cause skin & eye irritation, wear gloves & goggles when handling**

**® Benedict's soln is corrosive**

**® glucose is skin irritant**

**® Bunsen flame may cause burns thus handle with gloves. Flames should not be handled.**

**® hazard not stated in procedure**

[illegible]

### Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

- 5 (a)** Patient's own stem cells are frequently modified for use in treating genetic diseases via gene therapy.

Explain why these stem cells are suited for this purpose.

[6]

- 1. patient's own cells are used to prevent immune rejection**  
*as WBCs would not recognise these cells as foreign due to absence of Ag on cell surface;*
- 2. patient will not need to take immunosuppressant & immune sys ⊗ be compromised**  
*patient less susceptible to opportunistic infections;*
- 3. undifferentiated / unspecialised**  
*i.e. they ⊗ possess any specific cellular features;*
- 4. and are multipotent**  
*able to differentiate to give rise to limited cell types;*
- 5. able to undergo asymmetrical division to provide a supply of ASC**  
*and progenitor cells to differentiate into a related grp of cells to replace diseased cells;*
- 6. unlimited self-renewal capacity**  
*due to ✓ active telomerase that prevents telomere shortening with every cell division;*
- 7. preventing the cell from undergoing apoptosis due to replicative senescence;**
- 8. provides a supply of normal cells throughout lifetime of patient**  
*eliminating the need for repeated treatment;*

- (b)** The therapeutic gene used for gene therapy can be amplified using polymerase chain reaction (PCR). Describe the process of PCR. [8]

- 1. target DNA added to reaction tube with dNTPs, excess fwd & rev primers, Taq DNA pol in buffer**  
*in thermal cycler;*
- 2. denaturation**  
*heated to 95 °C;*
- 3. ↑ KE to break H bonds btw compl bp**  
*to separate ds DNA → 2 ssDNA;*



**4. annealing**

*lower temp to 55 °C;*

**5. to allow forward & reverse primers to bind**

*to seq flanking the target seq to provide free 3'OH end for Taq pol;*

**6. primers added in excess to ↑ chance of primer binding cf to DNA renaturing;**

**7. elongation**

*heating to 72 °C;*

**8. Taq pol binds to 3' end of primer to add DNA nucleotides w bases compl to target DNA / using target DNA as template;**

**9. syn new DNA strand in 5' → 3' direction;**

**10. cycle is repeated 25 – 30 times**

*to produce 2n copies of target DNA where n = no. of cycles;*

- (c)** The therapeutic gene can be delivered using viral or non-viral vectors. Compare the use of these 2 modes of gene delivery. [6]

**1. cf gene size;**

**2. cf tissue specificity;**

**3. cf ability to integrate thus stability of gene propagation;**

**4. cf possibility of insertional mutagenesis;**

**5. cf probability of triggering immune response;**

**6. cf ease of preparation;**

**Any elaboration on the answers to this?**

[Total: 20]

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